

THE RESULTS OF YOUR ANALYSIS

PREPARED FOR

Sylvia K

DATE OF IMAGES : 7/27/2016

DATE OF ANALYSIS : 7/27/2016

REFERRING DOCTOR :

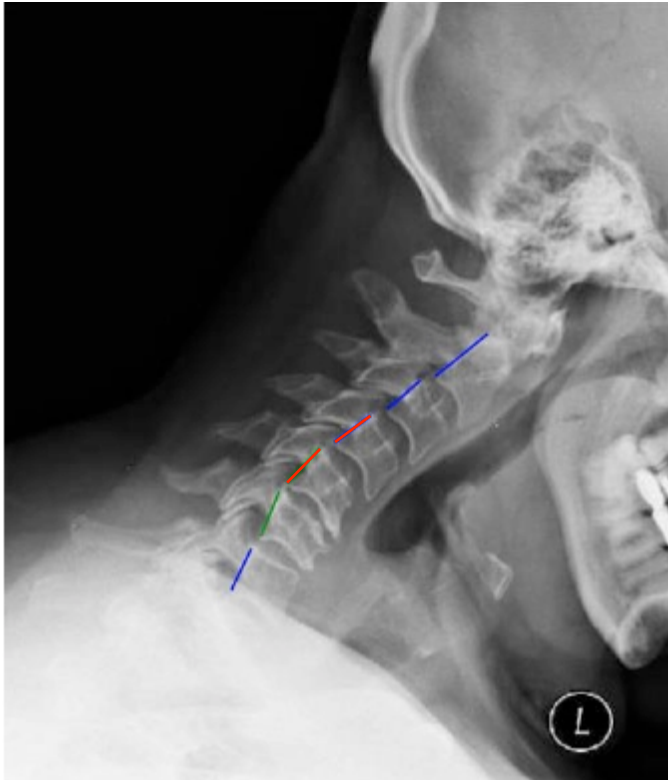
Dr. n

This report contains important information concerning structural changes that can be affecting your overall level of health and well-being. Spinal stability is a basic requirement for the protection of your nervous structures and the prevention of early mechanical deterioration of your spinal component

Instability is generally considered to be a global increase in the movements associated with the occurrence of back, neck, and/or nerve root pain. Damage to any spinal structure produces some degree of spinal instability.

This computer aided digital analysis is an overview of your current level of spinal stability.

PATIENT RESULT



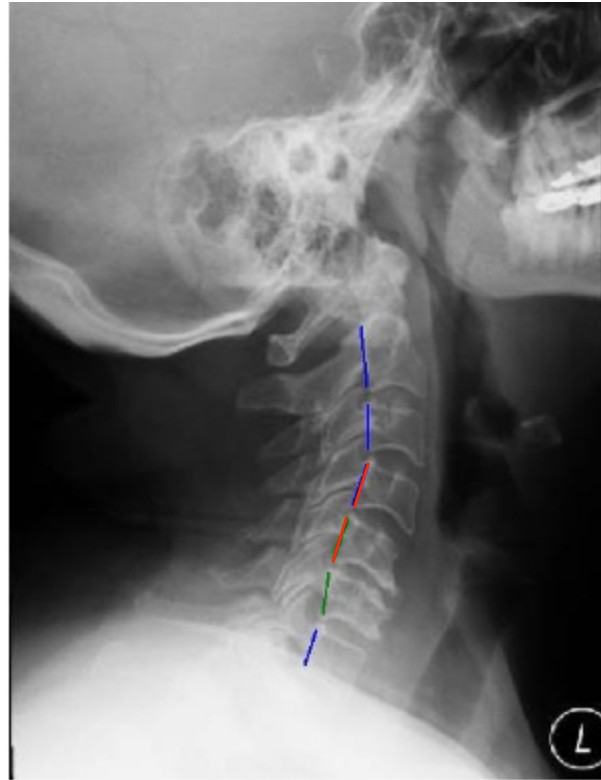
RED line denotes the location of the posterior longitudinal ligament (PLL) and surpasses normal acceptable segmental motion indicating the location of posterior longitudinal ligament laxity.

GREEN line denotes the location of the posterior longitudinal ligament (PLL) with normal alignment to the surrounding vertebrae indicating no significant ligamentous laxity.

Blue line denotes a segmental motion that is clinically significant.

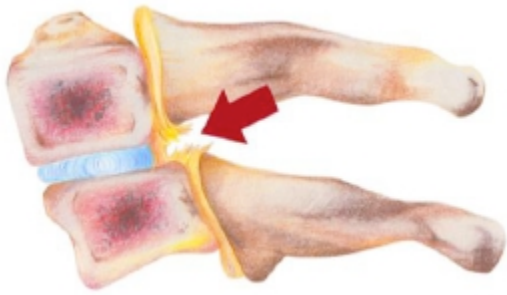
Alteration of motion segment integrity is defined as abnormal translation or angular motion between two adjacent vertebrae. This motion is measured using lateral flexion and extension x-rays. Translational motion segment integrity is assessed on flexion and extension imaging by comparing and measuring the position of the posterior edge of adjacent vertebral bodies in each position. Alteration of angular motion segment integrity is assessed by comparing the angles formed by the superior endplates of adjacent vertebral bodies (adjacent motion segments). These spinal translation and angulation are measured using a process referred to as Computed Radiographic Mensuration Analysis (CRMA). SpineTech Pro software captures articular motion to quantify spinal motion pathology and identify the presence and location of ligamentous laxity (ICD-9 728.4). The World Health Organization has rated this method of analysis as "ESTABLISHED".

PATIENT RESULT



Cervical Flexion / Extension - Translational Motion Results

Posterior Longitudinal Ligament Injury



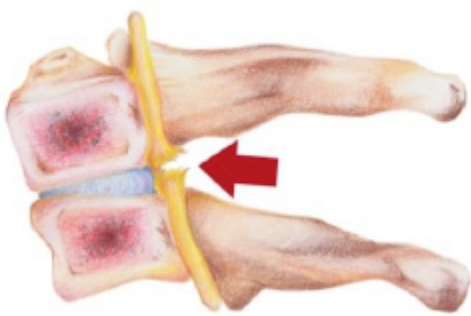
Anterior Longitudinal Ligament Injury



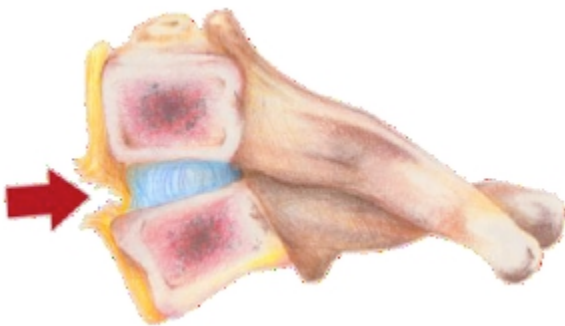
Vertebral Level	Flexion	Extension	Patient Result	Clinically		Comments
				Significant	Rateable Limit	
C2-C3	1.02	0.94	1.96 mm	1-3.5 mm	> 3.5 mm	Clinically Significant
C3-C4	0.54	1.1	1.64 mm	1-3.5 mm	> 3.5 mm	Clinically Significant
C4-C5	1.26	0.58	1.84 mm	1-3.5 mm	> 3.5 mm	Clinically Significant
C5-C6	0.16	0.32	0.48 mm	1-3.5 mm	> 3.5 mm	Normal
C6-C7	0.69	0.27	0.96 mm	1-3.5 mm	> 3.5 mm	Normal

Cervical Flexion / Extension - Angular Motion Results

FLEXION Posterior Longitudinal Ligament Injury



EXTENSION Anterior Longitudinal Ligament Injury



Vertebral Level	Flexion	Clinically Significant	Rateable Limit	Comments
C2-C3	0.66	7-11°	> 11°	Normal
C3-C4	3.5	7-11°	> 11°	Normal
C4-C5	11.84	7-11°	> 11°	Loss of Motion Segment Integrity
C5-C6	19.25	7-11°	> 11°	Loss of Motion Segment Integrity
C6-C7	3.44	7-11°	> 11°	Normal

Analysis Notes:

Measuring Translation :

in any of the spinal segments, lateral flexion and extension x-rays are analyzed. in the presence of true loss of motion segment integrity, the upper vertebrae will move forward on the lower vertebrae in flexion and move backwards on the lower vertebra in extension. Thus, the line along the back cortex of the upper vertebra will be anterior to the line along the posterior cortex of the lower vertebra on the flexion x-ray and posterior to it on the extension x-ray. The measurable angle is the sum of the two measurements.

If the total of the two measurements exceeds 4.5 mm in the lumbar spine or 3.5 mm in the cervical spine, the definition of loss of motion segment integrity is met. In the event that the line along the posterior cortex of the upper vertebra remains anterior to the line along the posterior cortex of the lower vertebra on the extension view, then consider only the distance between the lines on the flexion view, as it would not be appropriate to subtract the distance on the extension view from that on the flexion view.

Measuring Angular Motion:

Lateral flexion and extension x-rays are used. Lines are drawn along the superior cortex of the two vertebrae adjacent to the level in question on both the flexion and the extension x-rays and the angles where these lines intersect measured on both views. The angle on the extension view is then subtracted from the angle on the flexion view. Normally, the angle on the flexion view will be positive and the angle on the extension view will be negative. The net result will be that the negative extension angle will be subtracted from the positive flexion angle resulting in a net addition of the two angles. ($+8 - (-18) = +26$). If the angle is greater than 150 at L1-2, L2-3 or L3-4 or greater than 200 at L4-5 or greater than 250 at L5-S1, the definition of loss of motion segment integrity is met.

In the cervical spine, the method is different. A lateral flexion view only is taken. Lines are drawn along the inferior cortex of the two vertebrae above and below the level in question. Three angles are measured, the angles at the levels above and below the level in question and the angle at the level in question. If the flexion angle at the level in question is more than 110 greater than either of the angles at the levels above or below, the definition of loss of motion segment integrity is met. The figure in the 5th Edition is correct, but the text under the figure suggests that both flexion and extension films are necessary when actually only a flexion view is necessary.

These are the only definitions of loss of motion segment integrity.